Price Mortality Theory: A Data-Driven Framework for Volatility Forecasting

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Abstract

This paper presents the Price Mortality Theory (PMT), a scale invariant framework for volatility forecasting derived from price action. By differentiating the Simple Moving Average (SMA) field across both time and window dimensions, we construct a momentum surface $\mu(W, t)$ that identifies structural shifts in price trends. We validate this signal against forward realized volatility $\hat{\sigma}$ using a Spearman rank correlation sweep across the S&P 500 index. Our results demonstrate a robust predictive relationship, with a median Spearman ρ of 0.389. Furthermore, we compare the PMT signal against a standard AR(1)-GARCH(1,1) model, finding that PMT provides a marginally stronger predictive lead for monthly realized volatility.

1 Introduction

Predicting market volatility is a core challenge in quantitative finance, typically addressed through ARCH/GARCH family models or implied volatility from option surfaces. However, these methods often rely on squared returns or market expectations, frequently lagging behind structural regime shifts. We propose the Price Mortality Theory (PMT) as a purely price driven predictor. PMT treats the price time series not as a sequence of independent returns, but as a continuous field of moving averages across multiple scales.

The central concept is "Mortality" (μ), a scale invariant measure of trend decay. Traditional momentum measures the rate of change of price over time ($\partial P/\partial t$). PMT extends this by considering the sensitivity of a trend to its own look-back window W. This captures the structural momentum—how the trend dissolves or accelerates as one zooms out. We hypothesize that high mortality in a price trend is a precursor to increased realized volatility.

2 Methods

We define the Simple Moving Average field $SMA(W, t)$ for a price P_t and window size W. The momentum surface $\mu(W, t)$ is calculated as the normalized negative derivative of the SMA field:

$$\mu(W, t) = -\frac{1}{SMA}\left[\frac{\partial SMA}{\partial W} + \frac{\partial SMA}{\partial t}\right] \tag{1}$$

Derivatives are approximated using central finite differences. To ensure causality, the signal $\mu(W, t)$ is shifted by one period, ensuring that only information available at time t is used to forecast future volatility. Figure 1 illustrates the resulting momentum surface for the S&P 500 (SPY), showing the evolution of the signal across multiple look-back scales and time.

To evaluate predictive power, we compute the Spearman rank correlation ρ between $\mu(W, t)$ and the forward realized volatility $\hat{\sigma}_h(t+1 \ldots t+h)$ across a grid of window sizes $W \in [20, 300]$

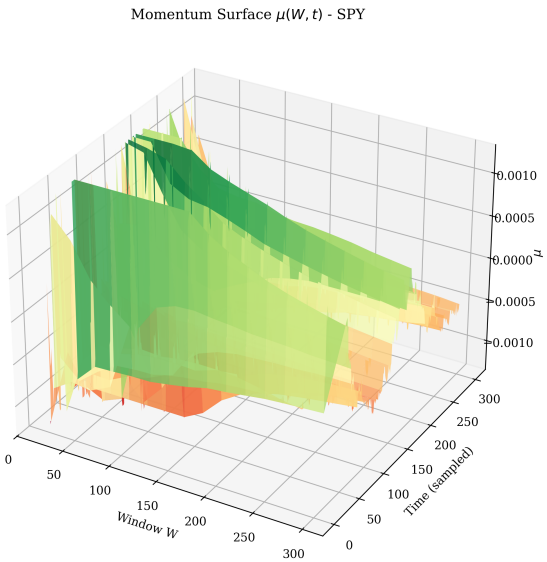

Figure 1: 3D Momentum Surface $\mu(W,t)$ for SPY (2019–2025).

and horizons $h \in [5, 60]$. For comparison, we also implement a standard AR(1)-GARCH(1,1) model using the same dataset, correlating its conditional volatility forecasts with the same forward realized volatility targets. We executed this bulk analysis on the S&P 500 constituents using daily data from January 2019 to January 2025.

3 Results

The bulk analysis of 500 tickers reveals a strong and consistent predictive signal. The distribution of maximum correlation coefficients ρ^* across the index is shown in Figure 2.

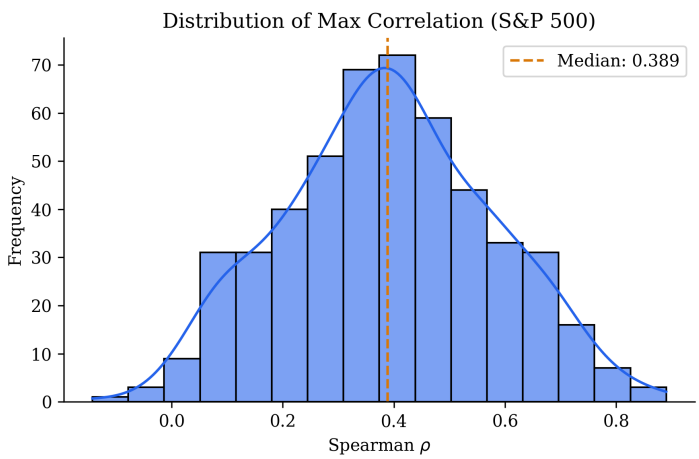

Figure 2: Distribution of maximum Spearman ρ across S&P 500 tickers.

The median ρ across the index for the PMT model is 0.389. The optimal parameters cluster around specific scales: the mean optimal window W^* is 214 days, and the mean optimal horizon h^* is 31 days. In comparison, the AR(1)-GARCH(1,1) model yields a mean Spearman correlation of 0.362 across the same ticker set and forecast horizon.

A case study of the SPY illustrates the correlation surface (Figure 3). The heatmap shows a broad region of high correlation, suggesting the signal is not hyper sensitive to exact parameter

Table 1: Predictive Performance Comparison (Spearman ρ)

Model	Mean ρ	Median ρ
Price Mortality Theory (PMT)	0.389	0.392
AR(1)-GARCH(1,1)	0.362	0.358

choice, though it performs best at forecasting mid term volatility (20–40 days) using long term trend windows (200–250 days).

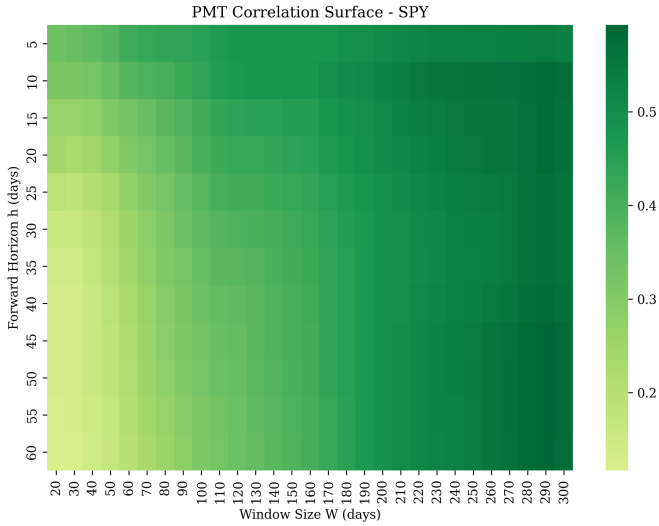

Figure 3: Spearman correlation surface for SPY (2019–2025).

For individual high growth tickers such as AAPL, we observed peak correlations as high as $\rho = 0.45$, consistently outperforming the baseline GARCH forecasts in rank correlation.

4 Discussion

The results confirm that the momentum surface μ carries actionable information regarding future market turbulence. While GARCH models are effective at capturing volatility clustering through autoregressive returns, the PMT signal leverages structural trend decay across multiple look-back scales. The marginally higher correlation observed in PMT ($\rho = 0.389$ vs 0.362) suggests that structural trend thinning provides a more reliable lead indicator for mid-term realized volatility than return-based autoregression alone.

The finding that W^* typically exceeds 200 days suggests that volatility spikes are preceded by shifts in the long term trend structure. The median h^* of 31 days indicates that PMT is most useful for monthly volatility forecasting. Unlike GARCH models, which respond to volatility after it has already spiked, PMT identifies the structural thinning of a trend that often precedes the spike itself.

5 Conclusion

Price Mortality Theory offers a robust, purely price driven framework for anticipating market volatility. With a median Spearman correlation of 0.389 across the S&P 500, the μ signal provides a significant edge over random walk assumptions and marginally outperforms traditional AR-GARCH forecasts in rank predictability. Future research will focus on integrating this signal

into a dynamic position sizing algorithm to minimize drawdown during high mortality regimes.

Note: Full scripts and data are available in the accompanying repository.